

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – ANSWERS

Course: Artificial Intelligence (CSA17)

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CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution and chaining methods.

CASE STUDY 1 – Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let F = Fever, C = Cough, R = Rash, B = Breathlessness, Flu = Possible Flu, M = Possible Measles, and P = Possible Pneumonia.

Rules: (1) $F \wedge C \rightarrow \text{Flu}$. (2) $F \wedge R \rightarrow M$. (3) $C \wedge B \rightarrow P$. Facts for Patient A: F = True, C = True, B = True.

(b) Forward Chaining

- Initial facts: Fever(A), Cough(A), Breathlessness(A).
- Rule 1 fires because Fever(A) and Cough(A) are true $\rightarrow$ derive PossibleFlu(A).
- Rule 2 does not fire because Rash(A) is not known.
- Rule 3 fires because Cough(A) and Breathlessness(A) are true $\rightarrow$ derive PossiblePneumonia(A).
- Final conclusions: PossibleFlu(A) and PossiblePneumonia(A).

(c) Backward Chaining for Pneumonia(A)

Goal: Pneumonia(A). The rule that concludes Pneumonia is $\text{Cough}(x) \wedge \text{Breathlessness}(x) \rightarrow \text{Pneumonia}(x)$. Reduce the goal to Cough(A) AND Breathlessness(A). Both are facts in the KB, so both subgoals are satisfied. Therefore Pneumonia(A) is proved.

CASE STUDY 2 – Autonomous Traffic Management Agent

(a) Unification

Rule 1: $\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$. For the fact $\text{Vehicle}(\text{Ambulance})$ and $\text{EmergencyType}(\text{Ambulance})$, the substitution is $\theta = \{x/\text{Ambulance}\}$. Therefore $\text{ClearPath}(\text{Ambulance})$ is derived.

Rule 2 with $x = \text{Ambulance}$ gives $\text{GreenSignal}(\text{Ambulance})$ and $\neg \text{RedSignal}(\text{Ambulance})$.

(b) Resolution

Relevant clauses: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$; $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$; $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$; $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$. Facts include $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$.

From Rule 1 and the ambulance facts: $\text{ClearPath}(\text{Ambulance})$. From Rule 2: $\text{GreenSignal}(\text{Ambulance})$. Substitute $x = \text{CarA}$, $y = \text{Ambulance}$ in Rule 3. With $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$, and $\text{GreenSignal}(\text{Ambulance})$, Rule 3 yields $\text{Proceed}(\text{CarA})$. Hence the goal is entailed.

(c) Two simultaneous emergency vehicles

The rules are universally quantified, so Rule 1 can independently derive ClearPath for each vehicle that has both Vehicle and EmergencyType facts. However, the KB lacks explicit rules for priority, conflict resolution, lane separation, or simultaneous green-signal coordination. Additional facts identifying each emergency vehicle and rules for prioritizing or safely coordinating multiple emergency routes are needed.

CASE STUDY 3 – Agricultural Expert Reasoning System

(a) CNF Conversion

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$ becomes $\neg\text{SoilDry} \vee \text{IrrigationNeeded}$.

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$ becomes $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$.

P3: $\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$ becomes $\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$.

Facts: SoilDry and CropWheat are unit clauses.

(b) Resolution Refutation for ApplyDripMethod

Negate the goal: $\neg\text{ApplyDripMethod}$. From $\neg\text{SoilDry} \vee \text{IrrigationNeeded}$ and SoilDry , derive IrrigationNeeded . Combine IrrigationNeeded , CropWheat and $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$ to derive ApplyDripMethod . Resolving ApplyDripMethod with the negated goal $\neg\text{ApplyDripMethod}$ produces the empty clause ■. Therefore ApplyDripMethod is proved.

(c) Removing SoilDry

Without SoilDry , P1 cannot derive IrrigationNeeded . Therefore P2 cannot derive ApplyDripMethod . However, CropWheat alone does not imply CropAtRisk because P3 also requires $\neg\text{ApplyDripMethod}$. Since $\neg\text{ApplyDripMethod}$ is not given or derivable from the remaining KB, CropAtRisk does not logically follow. Thus the conclusions become weaker and ApplyDripMethod cannot be established.

CASE STUDY 4 – Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Percepts: $\text{Stench}[1,2]$, $\text{Breeze}[1,1]$, $\text{Glitter}[2,2]$. From $\text{Stench}[1,2]$, derive $\text{WumpusAdjacent}[1,2]$. From $\text{Breeze}[1,1]$, derive $\text{PitAdjacent}[1,1]$. From $\text{Glitter}[2,2]$, derive $\text{Gold}[2,2]$. Rule 4 can derive $\text{Safe}[x,y]$ only when both $\neg\text{Wumpus}[x,y]$ and $\neg\text{Pit}[x,y]$ are known; such negative facts are not supplied, so no Safe cell can be concluded from Rule 4 alone.

(b) Forward Chaining – Possible Locations

The stench at $[1,2]$ indicates that the Wumpus is in a cell adjacent to $[1,2]$. The breeze at $[1,1]$ indicates that a pit is in a cell adjacent to $[1,1]$. The exact cell cannot be uniquely identified from the supplied rules because the KB only states adjacency, not a complete set of neighboring-cell constraints.

(c) Safety of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The path cannot be proven safe from the given KB. In particular, $\text{Stench}[1,2]$ provides evidence that a Wumpus is adjacent to $[1,2]$, and $\text{Breeze}[1,1]$ provides evidence of a nearby pit. Also, the Safe rule requires explicit negative knowledge that is not provided. Therefore the agent should not classify the entire path as logically safe based only on the supplied information.

Student Declaration

I, G Hema Chowdary (Register Number: 192473014), certify that this submission is my original work and follows the assessment guidelines.

Signature: _____